

EE 4345 – Semiconductor Electronics Design Project

RESISTORS

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OVERVIEW

- **RESISTANCE**
- **MEASUREMENT TECHNIQUES**
- **RESISTOR LAYOUT**
- **PROCESS VARIATION**
- **RESISTOR PARASITICS**

TYPES OF MATERIALS

- **CONDUCTORS**
- **SEMICONDUCTORS**
- **INSULATORS**

DEFINITION OF RESISTANCE

THE ABILITY WITH WHICH CURRENT FLOW IS ESTABLISHED AND MAINTAINED IS A METHOD OF CLASSIFYING MATERIALS AND IS COMMONLY REFERRED TO AS THE RESISTANCE OF THE MATERIAL.



SYMBOL

- **R**



UNITS

- **OHM (Ω)**



**ELECTRICAL
REPRESENTATION**



**MATHEMATICAL
REPRESENTATION**

- $R = (\rho * L) / A$

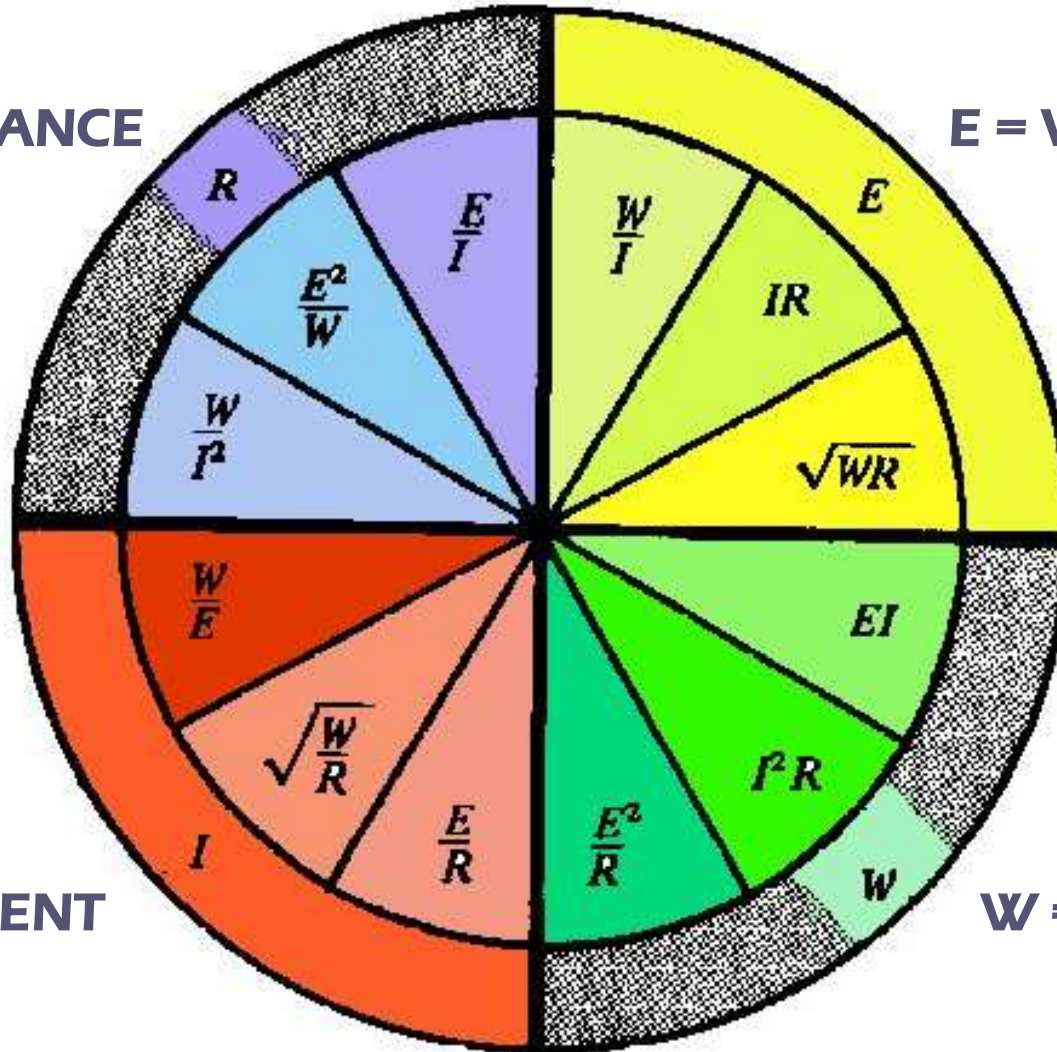
THE WHEEL SHOWS DC RELATIONSHIPS IN OHMS LAW

R = RESISTANCE

E = VOLTAGE

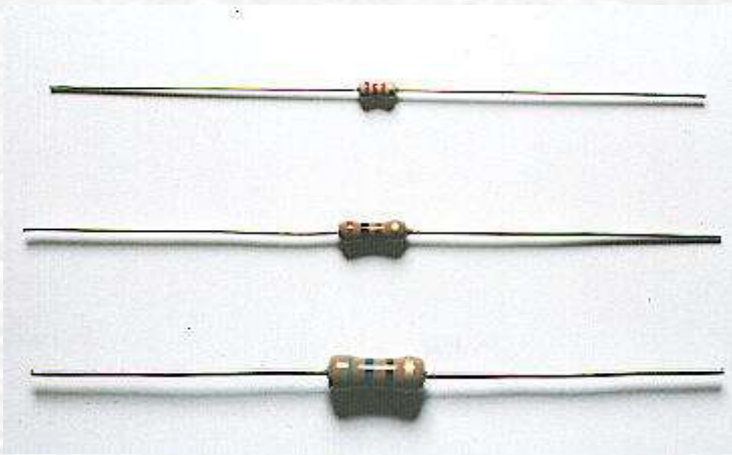
I = CURRENT

W = POWER



TYPES OF RESISTORS

CARBON FILM RESISTOR



SINGLE IN LINE RESISTOR NETWORK (SIL)



VARIABLE RESISTORS



THERMISTORS



RESISTOR COLOR CODES

BLACK=0

GREEN = 5

BROWN=1

BLUE = 6

RED = 2

VIOLET = 7

ORANGE = 3

GREY = 8

YELLOW = 4

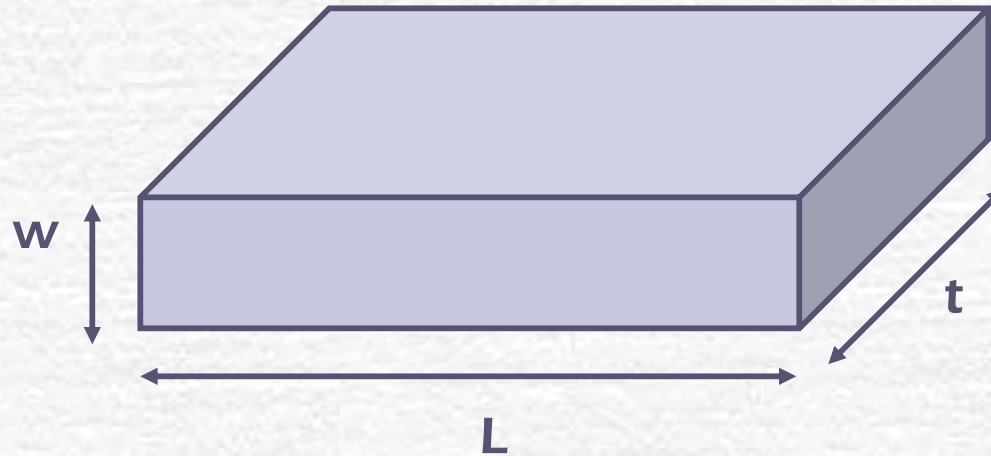
WHITE = 9

GOLD = 5 %

SILVER = 10%

**BAD BOOZE ROTS OUR YOUNG GUTS BUT
VODKA GOES WELL !!**

SHEET RESISTANCE (R_s)



$$R = \rho * L / (w * t)$$

$$R = R_s * L/w$$

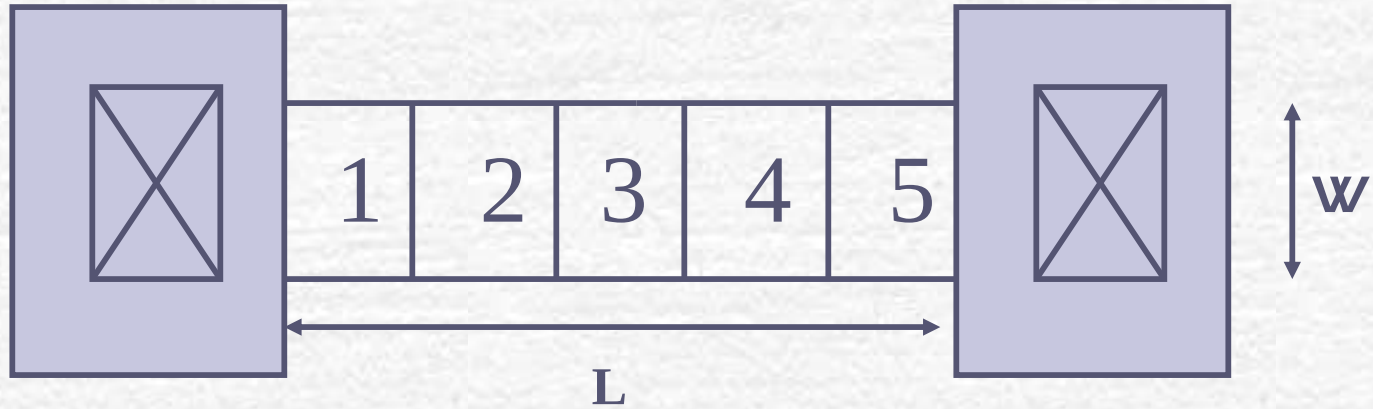
$$R_s = \rho / t$$

Units – Ohms per square ($\Omega / \square$)

EXAMPLE

CONTACT 1

CONTACT 2



$$L / W = 5$$

$$R_s = 50 \, \Omega / \square$$

$$R = R_s * L / W$$

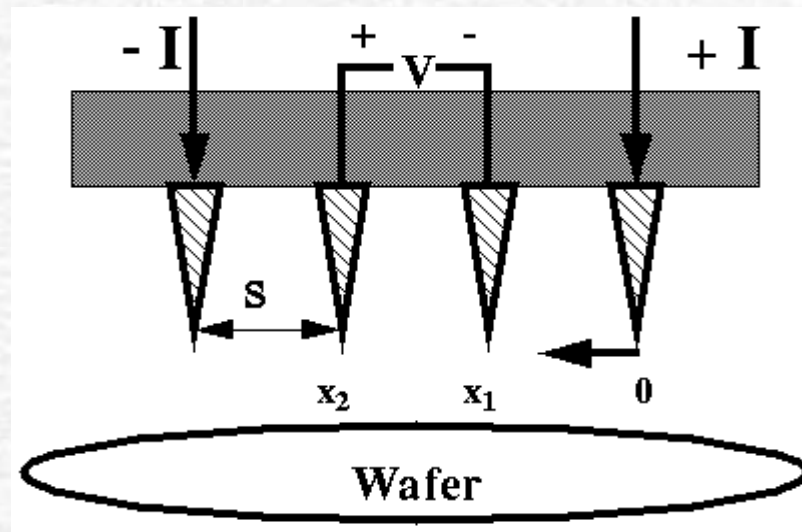
$$R = 250 \, \Omega$$

SHEET RESISTANCE MEASUREMENT

FOUR POINT PROBE

$$R_s = K * V / I$$

WHERE K = GEOMETRIC FACTOR





4 - POINT PROBE

**MODEL
FPP - 5000**

- **DIRECT CALCULATION OF V / I**
- **SHEET RESISTIVITY**
- **METALLIZATION THICKNESS**
- **P-N TYPE TESTING**



CPH - 2000 4 POINT PROBE

- **PORTABLE**
- **P/N TYPE SOUND REPORTING**
- **COMPUTERIZED ACCURACY**

WIDTH BIAS MODEL

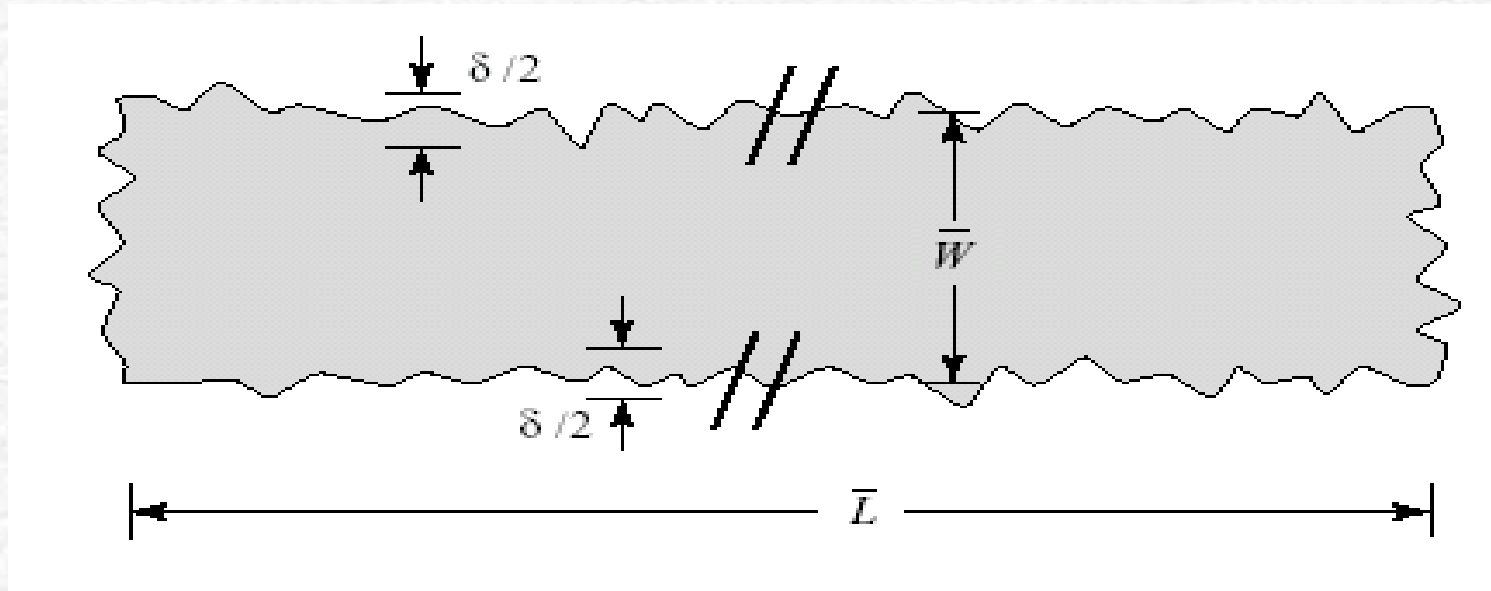


$$R = R_s * [L_d / (W_d + W_b)]$$

$$W_e = W_d + W_b$$

LINEWIDTH UNCERTAINTIES

- Due to lithographic and etching variation, the edges of a rectangle are “ragged”



$$W = \bar{W} (+/-) \delta$$

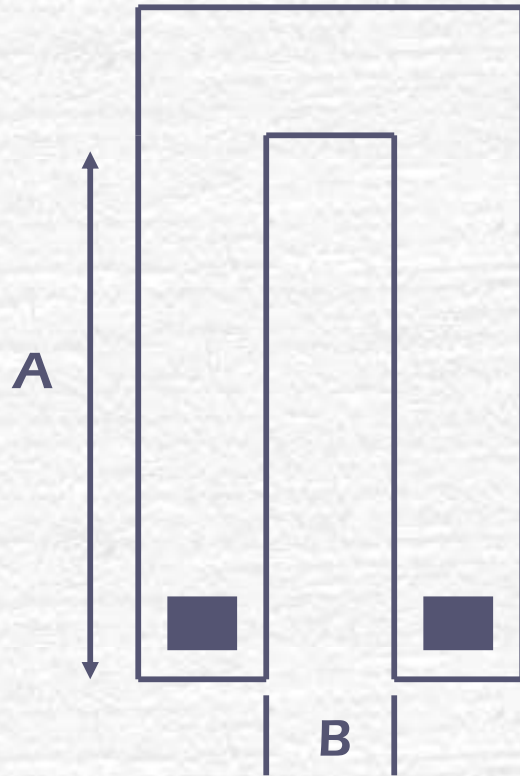
NON UNIFORM CURRENT FLOW

$$\Delta R = (R_s / \pi) * [(1/k) * \ln(k+1/(k-1)) + \ln((k^2-1)/k^2)]$$

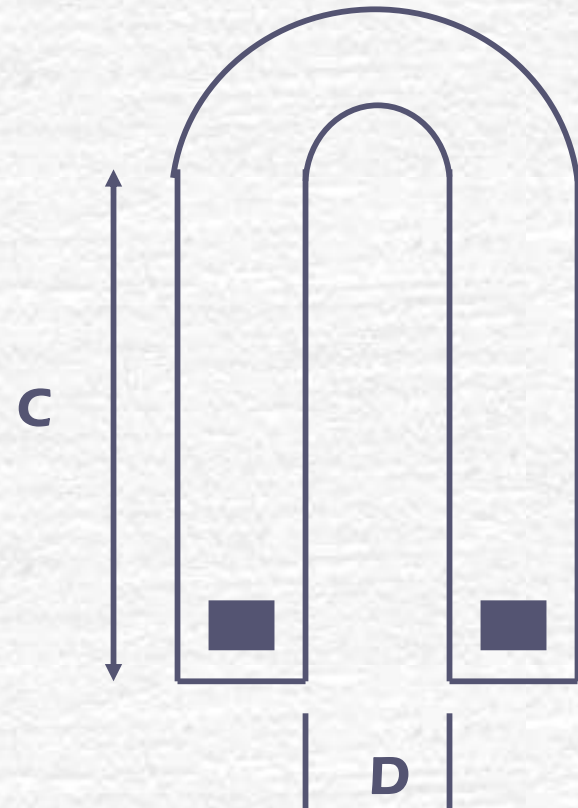
$$\text{where } k = W_e / (W_e - W_c)$$

ΔR represents the increase in resistance

SERPENTINE RESISTORS

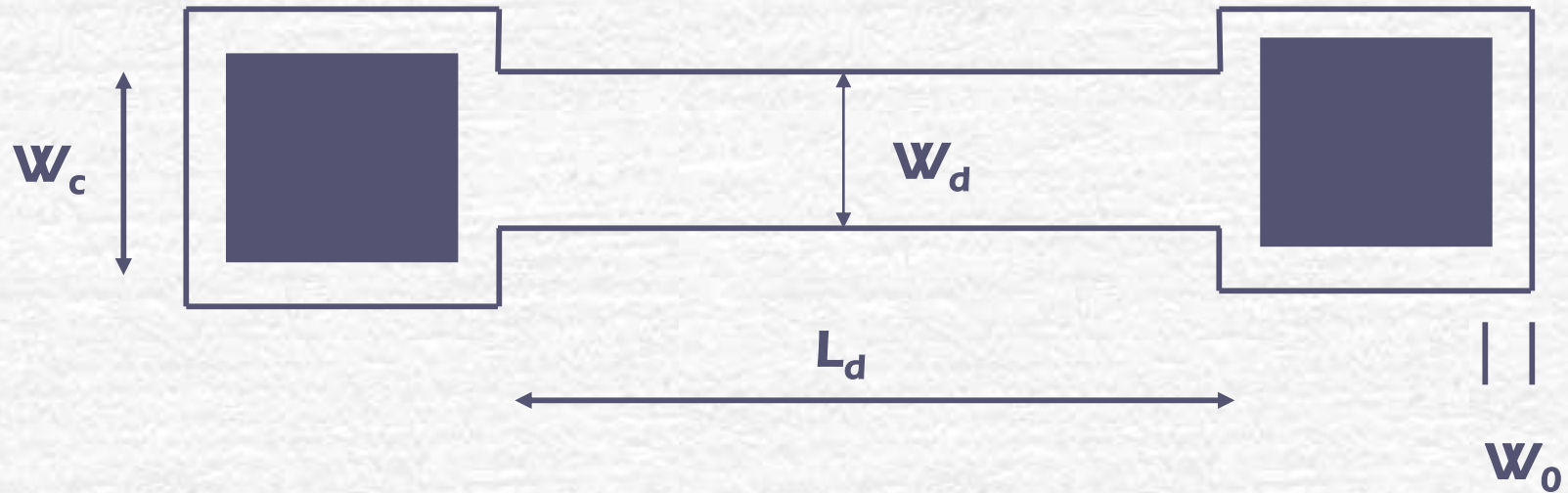


$$R = R_s(2A + B/W + 1.12)$$



$$R = R_s(2C/W + 2.96)$$

DOGBONE RESISTORS

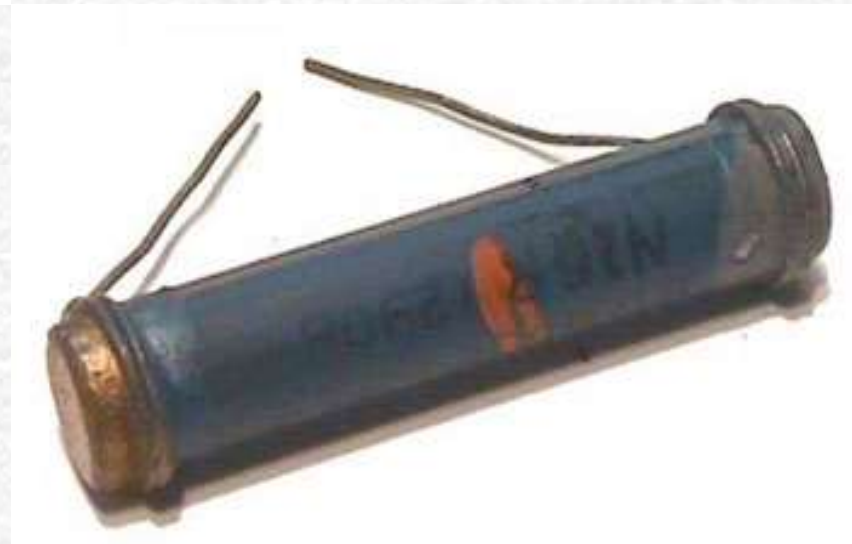


W_0	W_c	ΔR
W_d	W_d	-0.7 
$0.5 W_d$	W_d	-0.3 



RES-DBBNE-22/4

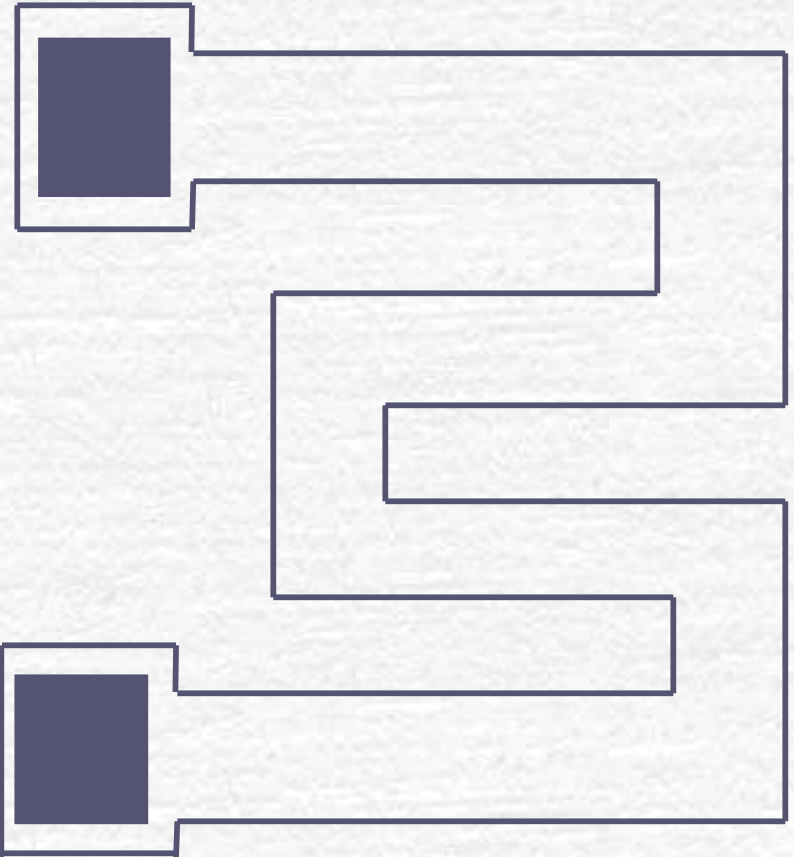
RES-DBBNE-100/4



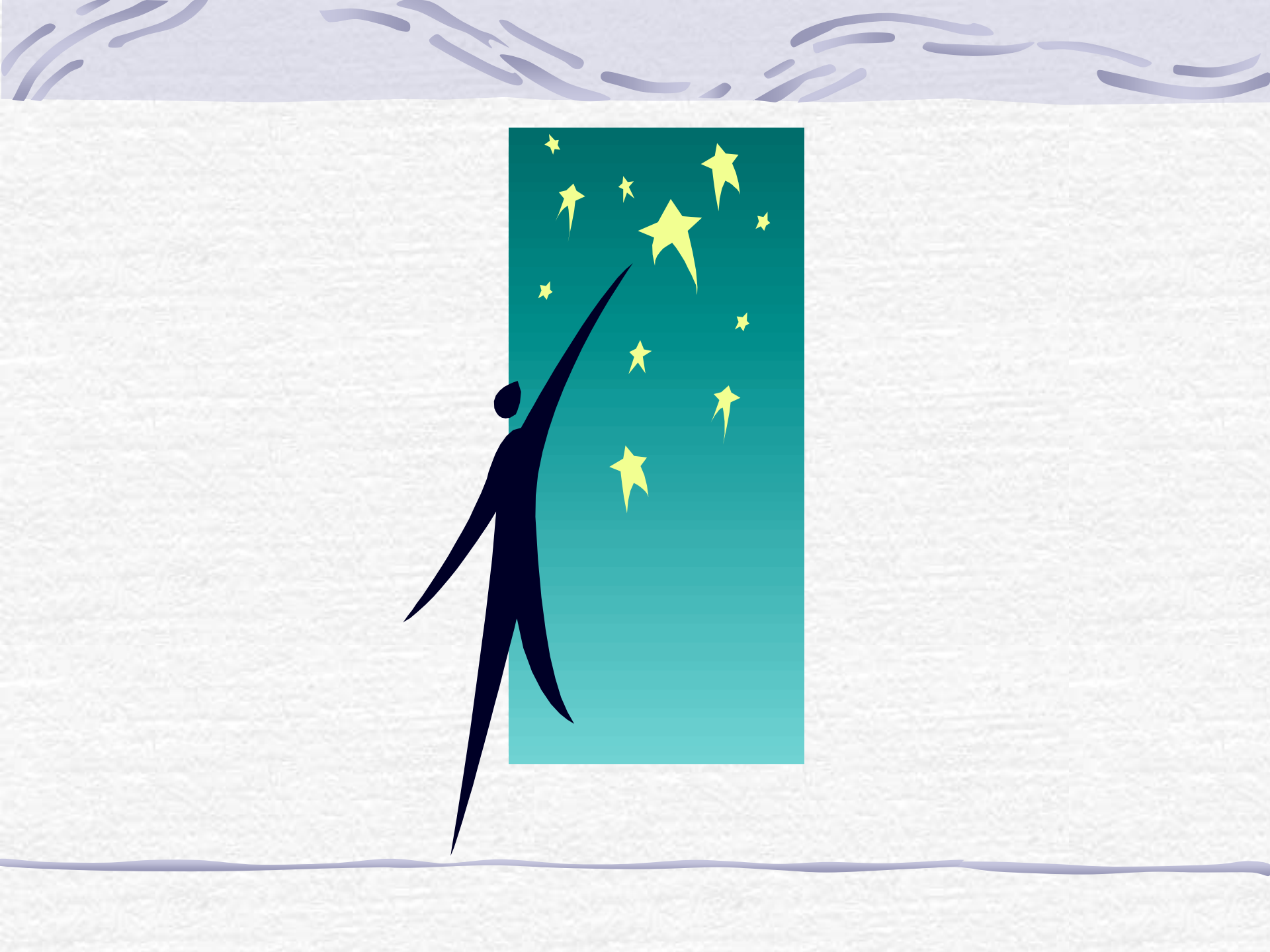
PACKING DENSITY



DOGBONE



SERPENTINE





RESISTOR VARIABILITY

THE VALUE OF A RESISTOR DEPENDS MAINLY ON THE FOLLOWING FACTORS :

- **PROCESS VARIABILITY**
 - **TEMPERATURE**
 - **NON-LINEARITY**
 - **CONTACT RESISTANCE**
- 

PROCESS VARIATION

$$R = R_s * L/W$$

where

R_s – SHEET RESISTANCE

FACTORS EFFECTING SHEET RESISTANCE

- FLUCTUATION IN FILM THICKNESS
- DOPING CONCENTRATION

**DIMENSIONS OF RESISTOR VARY BECAUSE OF
PHOTOLITHOGRAPHIC INACCURACIES**

ACTUAL TOLERANCE FOR A RESISTOR

$$\delta R = (C_L / W_E) + \delta R_s$$

where

δR – TOLERANCE OF THE RESISTOR

C_L – LINEWIDTH CONTROL OF THE
APPLICABLE LAYER

δR_s – VARIABILITY OF THE SHEET RESISTANCE

DESIGN GUIDELINES

- **WHERE TOLERANCE DOES NOT MATTER, USE MINIMUM WIDTH RESISTORS AND EXPECT VARIATIONS OF ABOUT + 50%**
- **WHERE MODERATELY PRECISE TOLERANCE IS REQUIRED, USE RESISTORS 2 TO 3 TIMES AS WIDE AS THE FEATURE SIZE AND EXPECT VARIATIONS OF + 35%**
- **WHERE MAXIMUM PRECISE TOLERANCE IS REQUIRED, USE RESISTORS 5 TIMES AS WIDE AS THE FEATURED SIZE AND EXPECT VARIATIONS OF + 30%**

TEMPERATURE VARIATION

RESISTIVITY DEPENDS ON TEMPERATURE IN A NON-LINEAR MANNER

$$R(T) = R(T_0)[1 + 10^{-6}TC_1(T - T_0)]$$

$R(T)$ - RESISTANCE AT THE DESIRED TEMPERATURE

$R(T_0)$ - RESISTANCE AT , ANOTHER TEMPERATURE, T_0

TC_1 - LINEAR TEMPERATURE CO-EFFICIENT OF RESISTIVITY IN PPM/°C

TYPICAL LINEAR TEMPERATURE COEFFICIENTS OF RESISTIVITY FOR SELECTED MATERIALS AT 25°C

MATERIAL	TCR PPM/°C
ALUMINUM	+3800
COPPER,BULK	+4000
GOLD,BULK	+3700
160Ω/□ BASE DIFFUSION	+1500
7Ω/□ EMMITER DIFFUSION	+ 600
5KΩ/□ BASE PINCH DIFFUSION	+2500
2KΩ/□ HSR IMPLANT (P-TYPE)	+3000
500Ω/□ POLYSILICON (4KÅ N-TYPE)	- 1000
25Ω/□ POLYSILICON (4KÅ N-TYPE)	+1000
10KΩ/□ N-WELL	+6000



NON-LINEARITY

FACTORS EFFECTING NONLINEARITY :

- **SELF HEATING**
 - **HIGH-FIELD VELOCITY SATURATION**
 - **DEPLETION REGION ENCROACHMENT**
- 

TEMPERATURE RISE BETWEEN THE RESISTOR AND THE SILICON SUBSTRATE IS GIVEN BY THE FOLLOWING EXPRESSION :

$$\Delta T = 71 * V^2 * T_{ox} / (R_s * L)$$

where

R_s – SHEET RESISTANCE OF THE POLY IN $\Omega/\square$

T_{ox} – THICKNESS OF THE FIELD OXIDE IN ANGSTROMS ($\text{\AA}$)

L - LENGTH OF THE RESISTOR IN MICRONS

V - VOLTAGE APPLIED ACROSS THE RESISTOR

THE MINIMUM RESISTOR LENGTH TO MINIMIZE NON-LINEARITY EQUALS

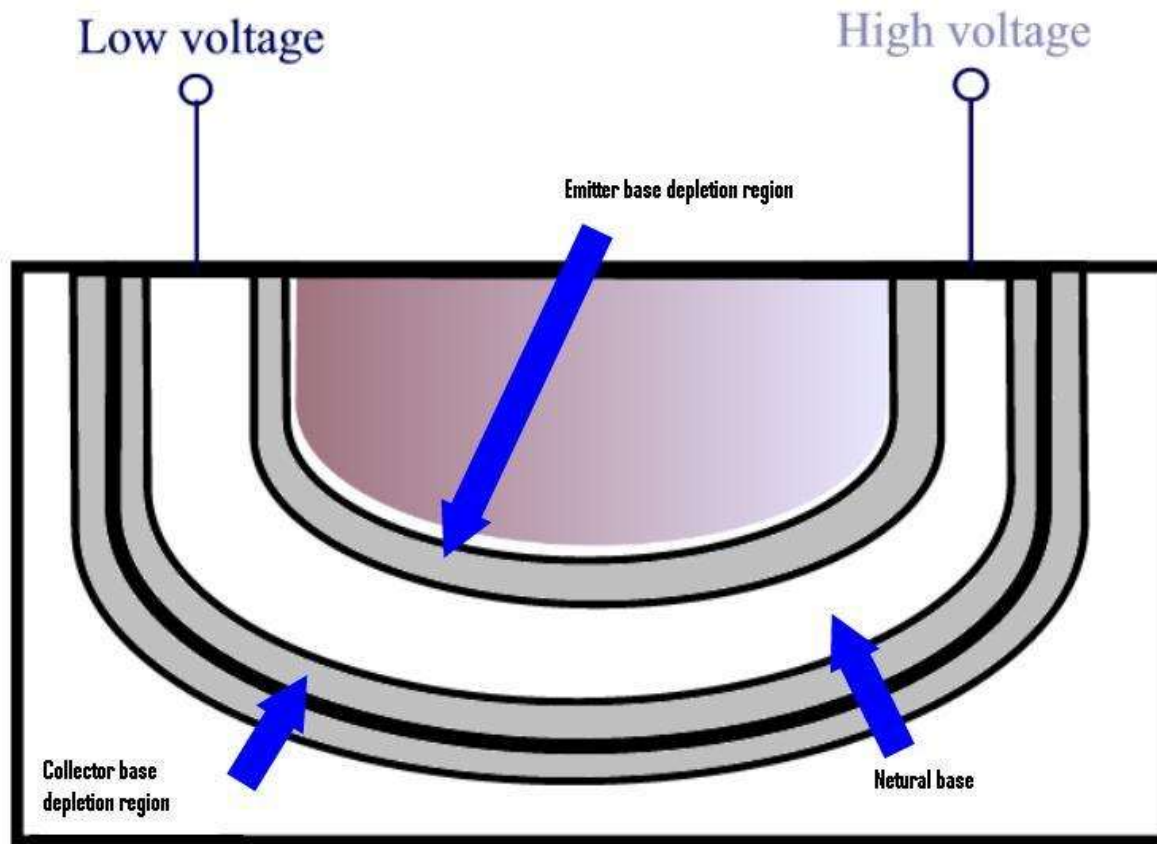
$$\text{L}_{\text{MIN}} = (6.7 \mu\text{M}/\text{V}) * \text{V}_{\text{MAX}} \quad \text{FOR N-TYPE SILICON}$$

$$\text{L}_{\text{MIN}} = (3.3 \mu\text{M}/\text{V}) * \text{V}_{\text{MAX}} \quad \text{FOR P-TYPE SILICON}$$

where

**V_{MAX} – MAXIMUM VOLTAGE APPLIED ACROSS
THE RESISTOR**

CROSS SECTION OF A BASE PINCH RESISTOR



TANK MODULATION

➤ **DEPLETION REGIONS CAUSE AN INCREASE IN RESISTANCE WHEN SIGNIFICANT TANK BIAS IS APPLIED.**

➤ **AS THE VOLTAGE DIFFERENCE BETWEEN THE RESISTOR AND THE TANK INCREASES, THE DEPLETION REGIONS WIDEN AND THE RESISTANCE INCREASES. THIS EFFECT IS CALLED TANK MODULATION.**

CONDUCTIVITY MODULATION

➤ CONDUCTIVITY MODULATION OCCURS WHEN THE ELECTRIC FIELDS GENERATED BY THE LEADS THAT CROSS A LIGHTLY DOPED RESISTOR CAUSE CARRIERS TO REDISTRIBUTE IN THE BODY OF THE RESISTOR.

CONTACT RESISTANCE

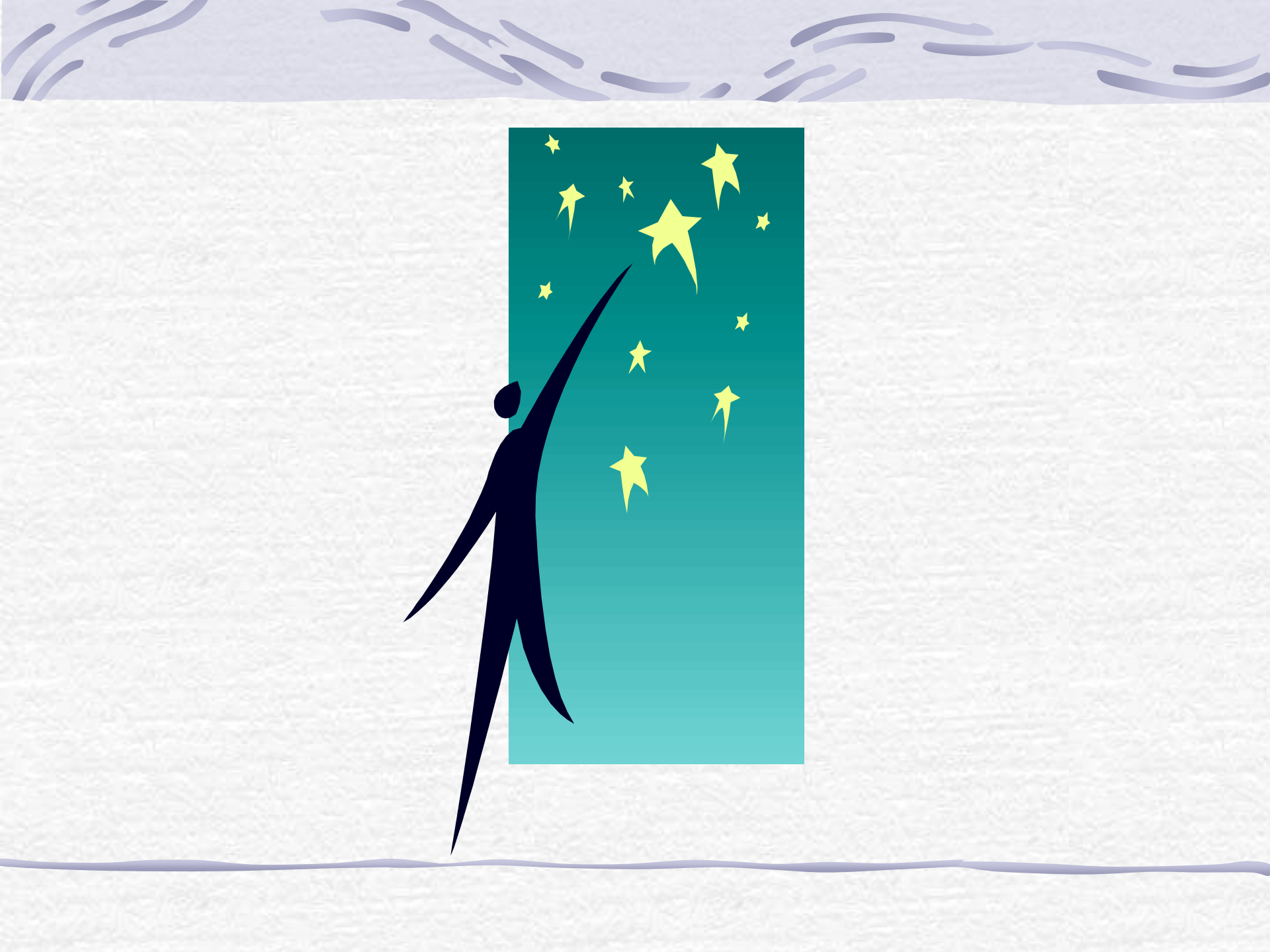
THE RESISTANCE R_C ADDED BY A SINGLE CONTACT HAVING WIDTH W_C AND LENGTH L_C EQUALS

$$R_C = (R_s * \rho_c)^{1/2} \text{COTH}(L_C * (R_s / \rho_c)^{1/2}) / W_C$$

R_s – SHEET RESISTANCE OF THE RESISTOR MATERIAL

ρ_c – SPECIFIC CONTACT RESISTANCE

$\text{COTH}()$ – IT REPRESENTS THE HYPERBOLIC COTANGENT FUNCTION

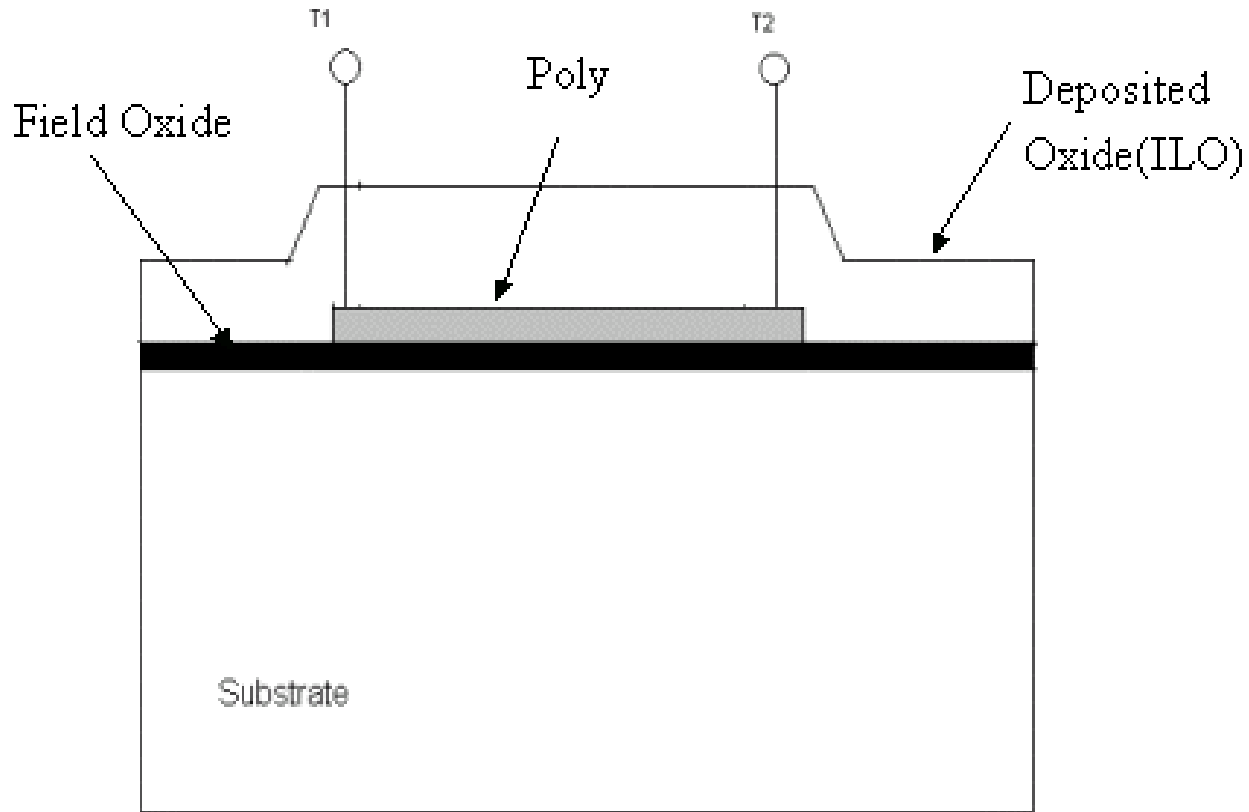




RESISTOR PARASITICS

- **CAPACITIVE AND INDUCTIVE COUPLING AT HIGH FREQUENCIES**
 - **JUNCTION LEAKAGE**
- 

POLYSILICON RESISTOR



CROSS SECTION OF POLYSILICON RESISTOR

CHARACTERISTICS OF OXIDE LAYER

- **INSULATOR PREVENTING LEAKAGE**
- **CAPACITIVE DIELECTRIC THAT COUPLES THE RESISTOR TO ADJOINING COMPONENTS**

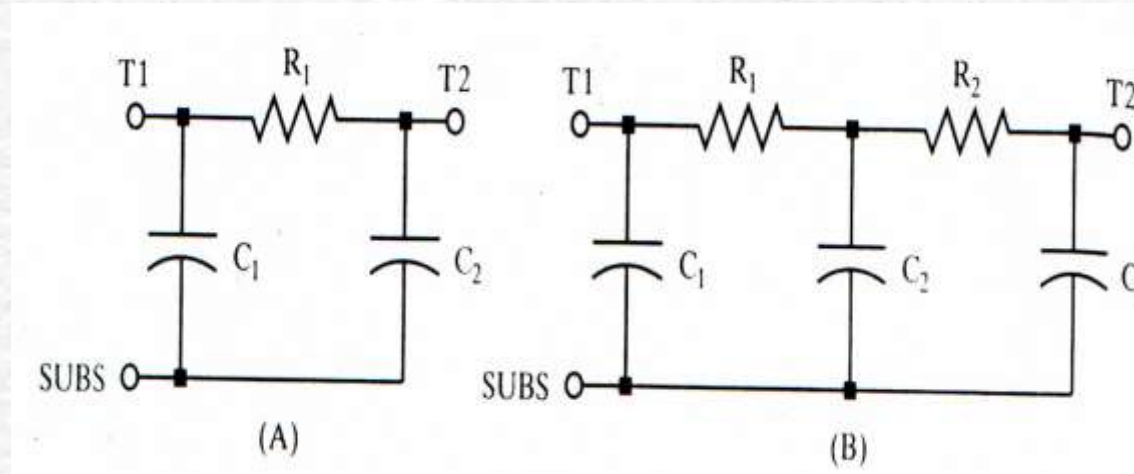
FIELD OXIDE LAYER

CAPACITANCE DUE TO FIELD OXIDE = $0.05 \text{ fF}/\mu\text{M}^2$

**CONSIDERING A RESISTOR OF $5 \mu\text{M}$ WIDE AND
CONTAINS 100 SQUARES,**

TOTAL SUBSTRATE CAPACITANCE = 0.125 pF

SUBCIRCUIT MODEL (π -SECTION)



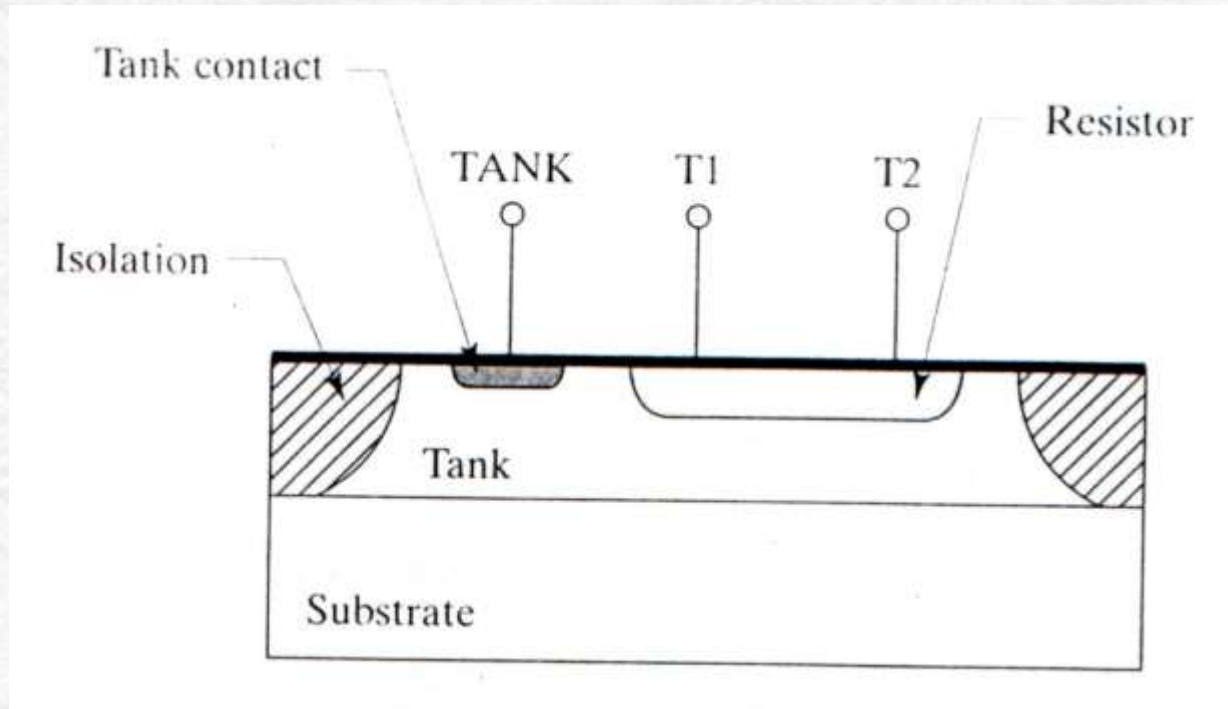
FOR TOTAL SUBSTRATE CAPACITANCE = C ,
IN FIG(A) $C_1 = C_2 = C/2$

IN FIG(B) $C_1 = C_3 = C/4$;
$C_2 = C/2$

INTERLEVEL OXIDE (ILO)

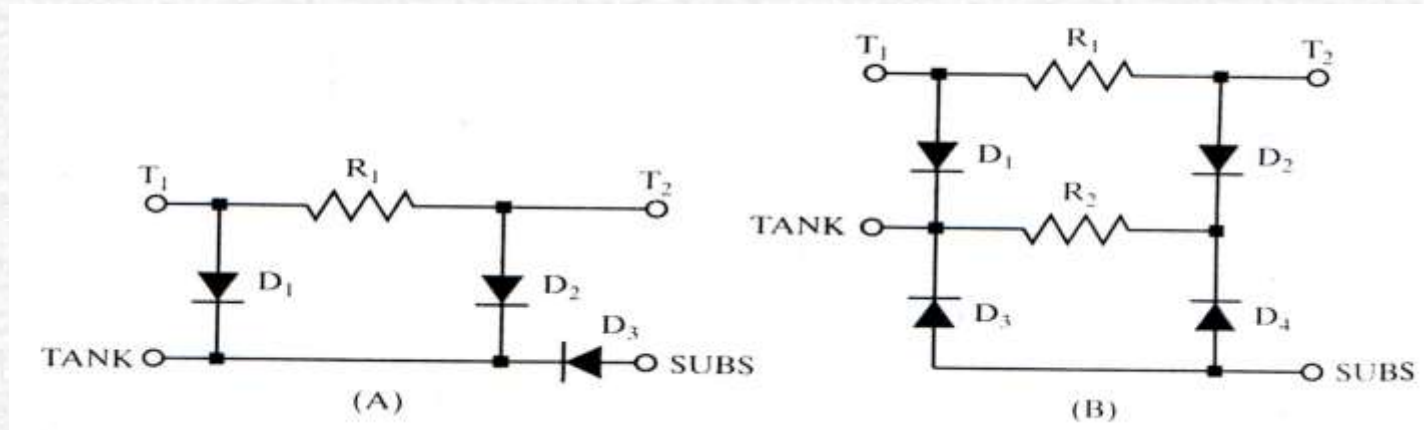
- **CAPACITANCE DUE TO ILO = $0.5 \text{ fF}/\mu\text{M}^2$**
- **CONSIDERING A $3 \mu\text{M}$ LEAD CROSSING A $5 \mu\text{M}$ –WIDE RESISTOR**
- **COUPLING CAPACITANCE = 7.5 fF**

DIFFUSED RESISTOR



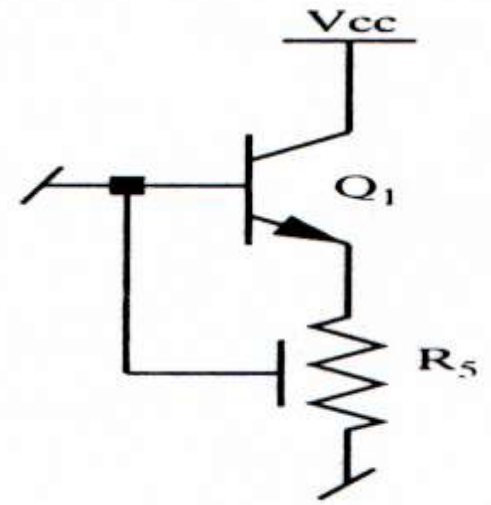
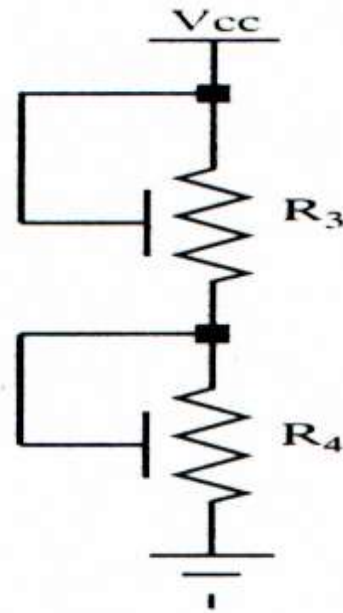
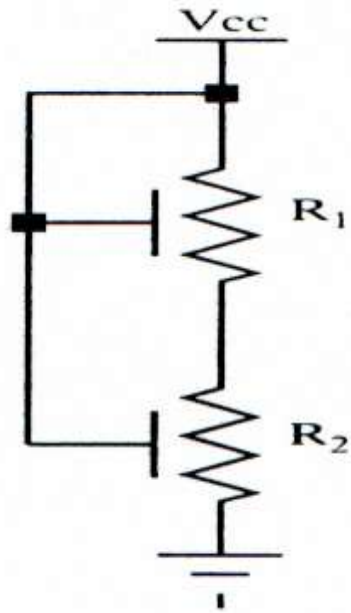
CROSS SECTION OF DIFFUSED RESISTOR

SUBCIRCUIT MODEL (π -SECTION)



- **IN FIGURE (A) FOR LOW TANK RESISTANCE:**
 - D_1 & D_2 = HALF OF THE TOTAL AREA OF RESISTOR-TANK JUNCTION
 - D_3 = FULL AREA OF TANK-SUBSTRATE JUNCTION
- **IN FIGURE (B) FOR HIGH TANK RESISTANCE:**
 - R_2 = TANK RESISTANCE
 - D_3 & D_4 = HALF OF THE TOTAL AREA OF TANK SUBSTRATE JUNCTION

TANK BIASING SCHEMES



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- **THE AVALANCHE BREAKDOWN IN THE REVERSE-BIASED JUNCTIONS OCCURS WHEN THE BIAS ACROSS A RESISTOR EXCEEDS ITS BREAKDOWN VOLTAGE. THIS CAN BE OVERCOME BY CONSTRUCTING MULTIPLE SEGMENTS IN SEPARATE TANKS.**

EXAMPLE: 7V FOR EMITTER RESISTOR AND BASE PINCH RESISTORS.

- **THE DEPLETION REGIONS ASSOCIATED WITH THE REVERSE-BIASED JUNCTIONS HAVE CONSIDERABLE CAPACITANCE DEPENDING ON THE DOPING AND THE REVERSE BIAS.**

EXAMPLE: TYPICALLY 1 TO 5 fF/ μm^2





REFERENCES

➤ **THE ART OF ANALOG LAYOUT BY
ALAN HASTINGS**

➤ **RESISTANCE AND RESISTORS BY
CHARLES WELLARD**

